

- (c) Chemical energy to heat energy
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

The energy change involves frictional force is kinetic energy to heat energy. When objects move against each other, frictional forces generate heat due to the microscopic interactions between the surfaces.

15. Which of the following makes use of Newton's third law of motion?

- (a) Archery
- (b) Space rocket
- (c) Venturimeter
- (d) More than one of the above
- (e) None of the above

B.P.S.C. School Teacher (Class 11-12) 15-12-2023

Ans. (d)

When the arrow is pulled back that is the action, the reaction is the arrow springing forward from spring force. A rocket engine produces thrust through action and reaction. The engine produces hot exhaust gases which flow out of the back of the engine. In reaction, a thrusting force is produced in the opposite reaction. Hence rocket and archery works on the Newton's third law of motion while Venturimeter works on Bernoulli's principle. So, option (d) is the correct answer.

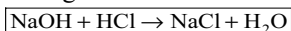
16. Lenz's law is derived from the law of conservation of

- (a) Magnetism
- (b) Momentum
- (c) Charge
- (d) More than one of the above
- (e) None of the above

BPSC (Tre-3) {Class 9-10}-2024

Ans. (e)

Lenz's law was given by the German Scientist Emil Lenz in 1834, this law is based on the principle of conservation of Energy.
According to Lenz's law—"The induced current in a circuit due to Electromagnetic Induction always opposes the change in Magnetic Flux.



Motion Under Gravity

1. If a feather, a rubber ball and a wooden ball are falling freely simultaneously from the same height in the vacuum then

- (a) The feather will reach the ground first
- (b) The rubber ball will reach the ground first

- (c) The wooden ball will reach the ground first
- (d) All three will reach the ground together
- (e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (d)

The feather, rubber ball and wooden ball will fall with the same speed and reach the ground together because there is no air resistance in the vacuum and gravitational force acts same on all the objects. Hence every object undergoes free fall in the vacuum with the same acceleration.

2. When a body falls freely towards the Earth, then its total energy

- (a) decreases
- (b) increases
- (c) remains constant
- (d) More than one of the above
- (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

When a body falls freely towards the earth, then its total energy remains constant because a freely falling body is free from the non-conservative and external forces, thus we can apply conservation of energy in such a case. By conservation of energy, the total energy of the body remains conserved, which is constant. As the body falls its potential energy starts converting into kinetic energy and the total energy remains the same.

3. A second pendulum is mounted in a rocket. Its period of oscillation decreases when the rocket

- (a) comes down with uniform acceleration
- (b) moves round the earth in a geostationary orbit
- (c) moves up with uniform acceleration
- (d) moves up with uniform velocity
- (e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (a)

As the rocket accelerates upwards, the pseudo force acts in the opposite direction of propagation.

Pseudo force acts in downward direction and gets added up to gravitational force.

Effective gravity = gravitational force + pseudo force

Gravitational force $g' > g$ where g' = effective gravity

Since time period of oscillation of pendulum = $\sqrt{L/g}$

where L = length of the pendulum

g = gravitational force acting on the pendulum

In this situation,

time period of oscillation of pendulum = $\sqrt{L/g'}$

Since $g' > g$

$\sqrt{L/g'} < \sqrt{L/g}$

Time period of oscillation decreases

4. For a body moving with constant speed in a horizontal circle, which of the following remains constant?

- (a) Velocity (b) Acceleration
(c) Centripetal force (d) Kinetic energy
(e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (d)

When a body moves at a constant speed in horizontal circle, its kinetic energy remain constant as mass and speed remain constant.

5. If the spinning speed of the earth increases, then the weight of the body at the equator will

- (a) increase (b) decrease
(c) remain same (d) be doubled
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (b)

The weight of a body on the Earth is the amount of gravitational pull on it exerted by the Earth. Due to Earth's rotation, centrifugal force is also exerted on a body by Earth, which tries to move the body away from the centre of the Earth. At the equator, centrifugal force is maximum as the velocity of rotation is maximum at the equator of the Earth. Thus, at the equator, centrifugal force acts against gravity to reduce the weight of the body. With the increase in the speed of rotation of the Earth, the centrifugal force at equator will increase and weight of the body will seem to have decreased.

6. The time period of a pendulum –

- (a) Depends on the mass (b) Depends on its length
(c) Depends on time (d) Depends on temperature

47th B.P.S.C. (Pre) 2005

Ans. (b)

The time period of a pendulum depends on its length. As we know that $T = 2\pi\sqrt{\frac{l}{g}}$, thus the result is that the one variable that affects the period of the pendulum is the length of the string. Increase in the length of string leads to increase in the period.

7. You are standing in an elevator. The force exerted by your weight on the floor of the elevator

- (a) will be zero when the lift goes up with an acceleration of 9.8 m/s^2 .
(b) will be zero when the lift comes down with an

acceleration of 9.8 m/s^2 .

(c) Zero when the lift moves in any direction with an acceleration of 9.8 m/s^2 .

(d) will not be affected by the change in the rate of acceleration of the lift.

46th B.P.S.C. (Pre) 2004

Ans. (b)

While going down from the lift with an acceleration of 9.8 m/s^2 , there is no pressure of the person on the floor of the lift, so the force acting by him on the floor at that time will be zero.

8. A bus is moving along a straight path and takes a sharp turn to the right side suddenly, the passengers in the bus will

- (a) bent towards right side
(b) fall in the forward direction
(c) bent towards left side
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

When a bus is moving straight and suddenly takes a sharp right turn, the passengers inside experience a curious phenomenon. As the bus turns, the cover portion of the passengers, those closest to the bus floor, is seen to be pulled towards the right side due to the centripetal force. On the other hand, the upper portion of the passengers seems to be thrown towards the left side of the bus due to the perceived centrifugal force.

9. A man pushes more weight on the Earth at

- (a) sitting position (b) standing position
(c) lying position (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Pressure exerted by a person on Earth =
= $\frac{\text{Weight of the person}}{\text{Surface area of the person covers}}$

$\left\{ \text{Pressure} = \frac{\text{Force}}{\text{Area}}, \quad \text{Force} = \text{Weight} \right\}$

A man can push higher weight on the earth when its covered surface area will lesser. Among the sitting, standing and lying position, the least area covered when a man is in standing position. Hence A man pushes more weight on the earth at standing position.